

Shiv Chandekar

Backend Engineer – Distributed Systems & Fintech Payments

📍 Mumbai, MH ✉ shivchandekar16@gmail.com ☎ +91-7767975374 in linkedin.com/in/shiv-chandekar
🌐 sscba.github.io

Summary

Java backend engineer with experience in building distributed, event-driven fintech systems at scale. Delivered 80% latency reduction, 18x throughput gain, and 70% cost savings on systems processing 70M+ transactions/day using Spring Boot, Kafka, Redis, and YugabyteDB.

Education

Indian Institute of Technology Bombay, Mumbai

Aug 2018 – Apr 2023

Bachelor of Technology in Metallurgical Engineering and Materials Science

Relevant Coursework: Data Structures, Algorithms, System Design, Design Patterns, HLD, LLD

Professional Experience

Software Engineer - MindGate Solutions Pvt. Ltd., Mumbai

Jul 2023 – Jan 2026

UPI Reconciliation System – Large-Scale Payment Processing Platform

- Architected a Java/Spring Boot distributed reconciliation system processing **70M+ UPI transactions/day** across 10 settlement cycles, with near-real-time discrepancy detection across Switch, NPCI, and CBS
- Built Kafka ingestion pipelines (**20+ partitions, 5K–20K+ events/sec**) into YugabyteDB (Cassandra API) with a write-optimized schema for high-throughput
- Engineered a Redis Streams + Redis Gears reconciliation engine running **2M+ checks/cycle**, reducing latency by 80%
- Designed distributed job orchestration and Quartz pipelines across **6+ microservices** using Redis, reducing manual reconciliation effort by **60%**

HyperStream Analytics POC – Cost-Optimized Transaction Processing

- Migrated **70M+ UPI records** from Spark-Kafka to DuckDB vectorized processing, cutting batch runtime by **97% (4+ hours to 15 minutes)** with zero data integrity loss
- Replaced a Spark-kafka microservices architecture with a single-node DuckDB setup, achieving **18x throughput** and **70% cost reduction**

Key Projects

Kafka Topology Visualizer with Spring AI

Apr 2026

- Engineered a production-grade **Spring AI MCP server** in Java 21 that exposes **5 real-time** Kafka analysis tools to Claude AI over a **stdio/JSON-RPC 2.0**, enabling natural-language cluster interrogation without REST APIs.
- Designed and implemented **5** Kafka diagnostic tools — **topology inspection, consumer lag tracking, message flow tracing, hot-partition detection, and AI-driven partition strategy recommendations** — queryable entirely through conversational Claude prompts against a live cluster.
- Reduced AdminClient round-trips by 2× by introducing Caffeine **TTL caching (30s for topology, 15s for consumer lag)**, eliminating redundant **Kafka metadata** calls on repeated **Claude tool invocations**.

TaskManagerAPI – Enterprise-Grade Task Management System

Jan 2026

- Built a Spring Boot REST API (**20+ endpoints**) with JWT auth, RBAC, rate limiting (100 req/min), and email verification, sustaining **under 200ms** response time under load
- Cut critical-path latency by **90% (5–6s to 500ms)** via Spring **@Async** email processing, BCrypt tuning, and targeted database indexing
- Achieved **80%+ unit test coverage** with JUnit/Mockito and **34 Postman integration tests**; documented via OpenAPI/Swagger with Actuator observability

Technical Skills

Programming Languages: Java, Python, C++, Lua, Bash

Backend Technologies: Spring Boot, Spring MVC, Spring AI, RESTful APIs, Microservices, MCP (Model Context Protocol), JSON-RPC 2.0, Kafka AdminClient API, SASL Authentication, Caffeine Cache

Databases & Storage: Oracle SQL, NoSQL, Redis, Yugabyte (PostgreSQL, Cassandra), DuckDB

Big Data & Streaming: Apache Kafka, Apache Spark, Data Pipelines, Redis Streams, Distributed Streaming Systems

Cloud & DevOps: Docker, Kubernetes, Linux, Git, Maven

Frontend & AI Technologies: Streamlit, ollama, LangChain, Anthropic Claude AI Integration, LLM Tool Calling

Testing & Documentation: Postman, Swagger, OpenAPI, JUnit, Mockito

Concepts: Distributed Systems, Event-Driven Architecture, System Design, Concurrency, High Availability, Agile